

have become obvious to you that the technology is not simple. At various levels, we are in need of different techniques to achieve the goal. In the following few pages, we talk about the technologies that bind together the various parts to produce the mesmerizing effects of AR. We would go through each type one after another.

Projection based AR

We start off with projection based AR because the number of technologies and complexities of their interrelationships are the minimum here. In case of projection based AR, one has to deal with only a few parameters. The primary point of concern, as we have already said, is the creation of image sequence which is to be projected. Depending on the needs, quite a number of tools can be used. Let us begin with non-interactive AR.

Non-interactive AR

If the projection surface is just a plane surface, it is easy to generate the video sequence. The software required for the same would depend, once again on the specific need. If the need is only images, basic image processing tools along with video editing tools suffice.

For more advanced uses which have captivating effects, more tools are required. Let us reconsider the case of projection on a cube with an edge-length of 2 meters. It is difficult to simply imagine a sequence which would look good on the cube without actually testing it. Let us not forget that the orientation of the cube matters – it may be kept on one of its sides or tilted on an edge or be placed on one of the corners. The variety of options depend on the projection surface and its complexity.

In such cases a 3D modelling and animation software can prove immensely useful. Most such software can simulate any 3D environment and allow the designer to find out the best image sequences. They are also useful when experimentation space is not large enough. Since the software can produce animation with 3D objects, they can also help in generation of the actual light sequence to be projected. 3D animation software in non-interactive projection based AR can serve as a WYSIWYG tool for the result.

Interactive AR

In cases where projection based AR is used for interactive purposes, the target surfaces are mostly planar and the projection is not a light sequence that needs to be designed with a video editor or 3D animation tool well before